

EXPERIMENT NO. 5

Aim: Apply array methods and object handling; create a cart total calculator with discount logic.

Requirements: A computer with a web browser, Visual Studio Code (or any text editor), HTML, CSS, and JavaScript.

Theory:

Arrays, Objects, and Cart Total Calculation in JavaScript

JavaScript provides powerful data structures such as **arrays** and **objects** for storing and managing data efficiently. Arrays allow multiple values to be stored in a single variable, while objects store data in the form of key-value pairs. JavaScript also offers various built-in array methods that simplify data manipulation and processing. These concepts can be applied to develop practical applications such as a shopping cart total calculator with discount logic.

1. Arrays in JavaScript

An array is a collection of multiple values stored under a single variable name. Each element in an array is identified by its index, starting from 0.

Arrays are useful for storing lists of related items and performing operations on them efficiently.

2. Array Methods

Some important array methods are:

- **push()** – adds an element at the end of an array.
- **pop()** – removes the last element.
- **shift()** – removes the first element.
- **unshift()** – adds an element at the beginning.
- **splice()** – adds, removes, or replaces elements at a specific position.
- **slice()** – extracts a portion of an array without modifying the original array.
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JavaScript also provides higher-order array methods:

- **map()** – creates a new array by applying a function to every element.
- **filter()** – creates a new array containing elements that satisfy a condition.
- **reduce()** – reduces an array to a single value.
- **forEach()** – executes a function for each element of an array.

Loops such as **for**, **while**, and **for...of** can also be used to traverse and process array elements.

Algorithm:

1. Start the program.
2. Create an array containing a set of numbers.
3. Display the original array.
4. Use array methods to manipulate or process the array.

5. Find the maximum value using `Math.max()`.
6. Find the minimum value using `Math.min()`.
7. Display the maximum and minimum values.
8. Stop the program.

Conclusion:

Arrays and objects are fundamental data structures in JavaScript that enable efficient storage and manipulation of data. Array methods such as `map()`, `filter()`, and `reduce()` simplify data processing tasks, while objects help organize related information. By combining these concepts, a cart total calculator can accurately calculate item totals, apply discounts, and generate the final payable amount, demonstrating practical use of JavaScript in real-world applications.

